# Lecture 12: Mesh Current Analysis

## EE201 – Circuit Theory I

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current, mesh current 1 voltage supermesh

# Review: node voltage analysis

## The procedure

- Ground; node voltages $\mathbf{e}$
- KVL automatic: $\mathbf{v} = A^T \mathbf{e}$
- KCL at non-ground nodes
- $Y_n \mathbf{e} = \mathbf{i}_n$

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- To the ground: $e_k$ known
- Between two nodes: supernode
- Dependent: one more constraint

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The opposite?

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# Motivation

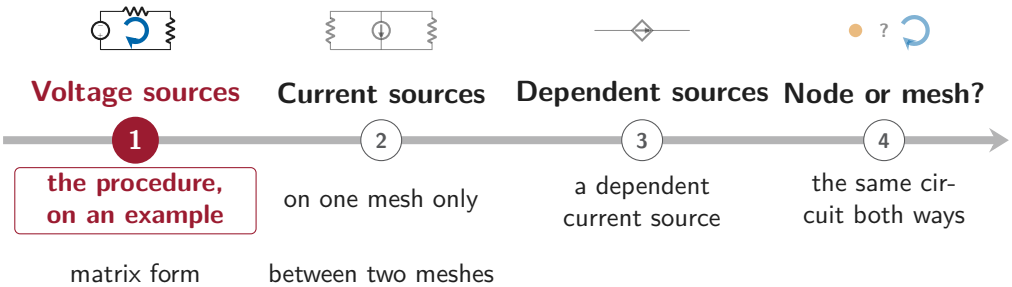
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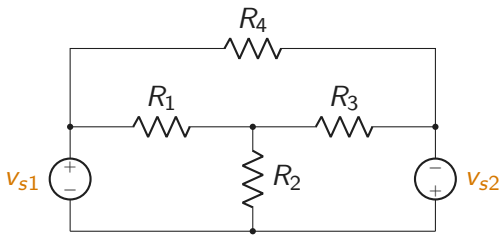
- Unknowns: $\ell = b - n + 1$ mesh currents
- KCL automatic: $\mathbf{i} = M^T \mathbf{i}_M$
- Write KVL
- Needs a planar circuit

# Roadmap


Voltage sources
Current sources
Dependent sources
Node or mesh?
1
2
3
4
the procedure, on an example
matrix form
on one mesh only
between two meshes
a dependent current source
the same circuit both ways

# Example 12.1: mesh current analysis step by step

For now: only resistors and independent voltage sources

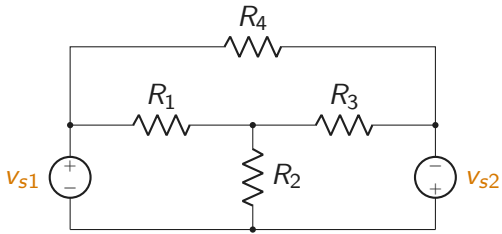



**Goal:** every branch current and voltage

**Four steps:**

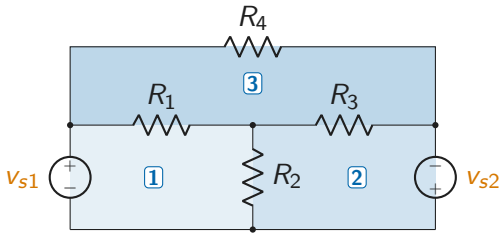
1. Find the inner meshes
2. Assign the mesh currents
3. Resistor voltages with mesh currents
4. KVL around every inner mesh

# Step 1: find the inner meshes




Planar: no crossing branches

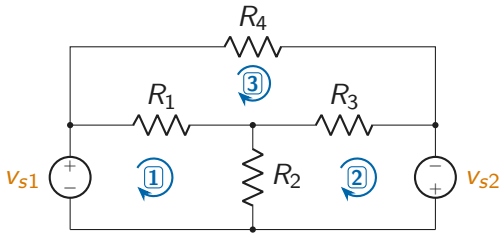
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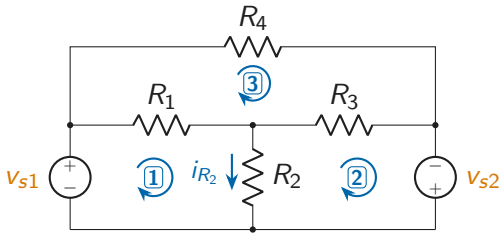
Inner meshes: the windows

# Step 2: assign the mesh currents




Unknowns, clockwise: $i_{M1}$, $i_{M2}$, $i_{M3}$

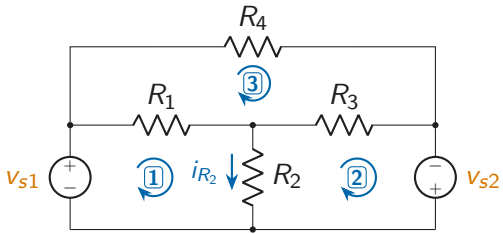
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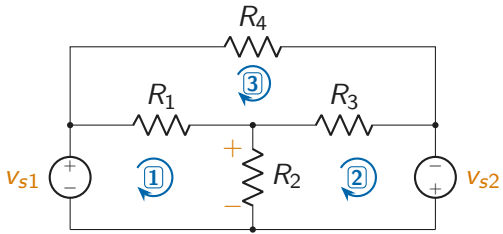



Unknowns, clockwise: $i_{M1}$, $i_{M2}$, $i_{M3}$

$i_{R_2} = i_{M1} - i_{M2}$

KCL holds automatically

# Step 3: write the resistor voltages with mesh currents



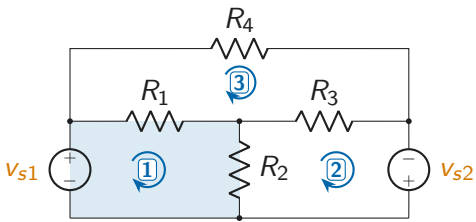

$R_2$, along mesh 1:

$v = R_2(i_{M1} - i_{M2})$

$R_4$: $v = R_4\, i_{M3}$

$R$ (own − other)

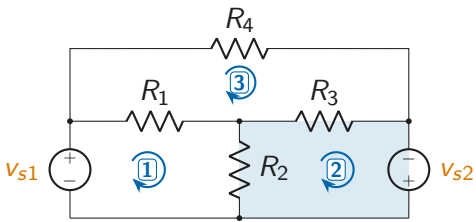
# Step 4: KVL around every inner mesh




**Mesh 1:**

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

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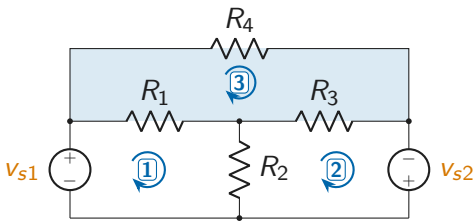

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$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

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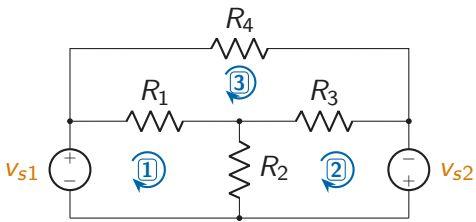
**Mesh 2:**

$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

**Mesh 3:**

$$R_4\, i_{M3} + R_3(i_{M3} - i_{M2}) + R_1(i_{M3} - i_{M1}) = 0$$

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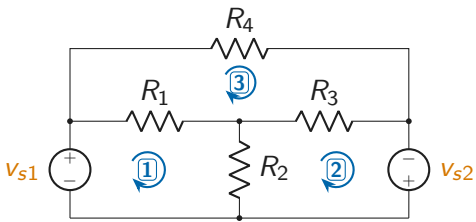
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Source met $-$ to $+$: minus

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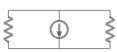
$$R_4\, i_{M3} + R_3(i_{M3} - i_{M2}) + R_1(i_{M3} - i_{M1}) = 0$$

Source met $-$ to $+$: minus

3 equations, 3 unknowns

# Roadmap









**Voltage sources** | **Current sources** | **Dependent sources** | **Node or mesh?**

1 — the procedure, on an example ✓

2 — on one mesh only

3 — a dependent current source

4 — the same circuit both ways

**matrix form** between two meshes

# Example 12.1 in matrix form

$$
\begin{aligned}
(R_1 + R_2)\, i_{M1} - R_2\, i_{M2} - R_1\, i_{M3} &= v_{s1} \\
-R_2\, i_{M1} + (R_2 + R_3)\, i_{M2} - R_3\, i_{M3} &= v_{s2} \\
-R_1\, i_{M1} - R_3\, i_{M2} + (R_1 + R_3 + R_4)\, i_{M3} &= 0
\end{aligned}
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One matrix equation?

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$Z_M$: mesh impedance matrix

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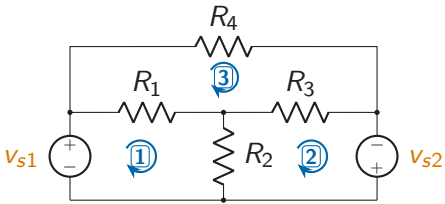
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$Z_M$: mesh impedance matrix

$\mathbf{v}_M$: source voltages rising along each mesh

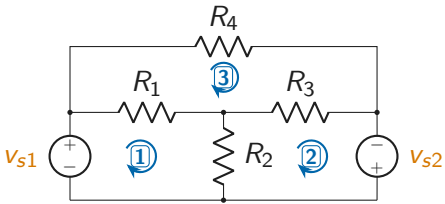
# Writing $Z_M$ directly from the circuit




Read $Z_M$ from the circuit?

$$\underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

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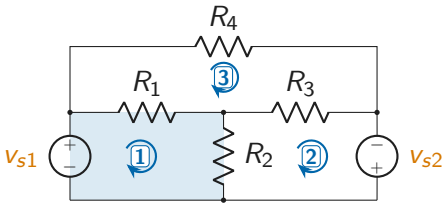



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**Diagonal** $(k, k)$: $+\sum R$ around mesh $k$

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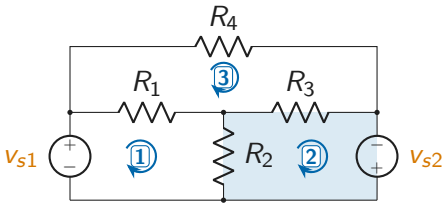



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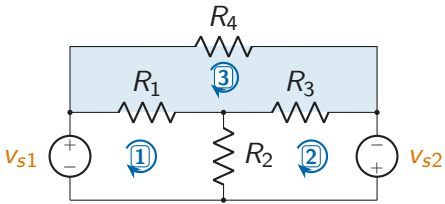



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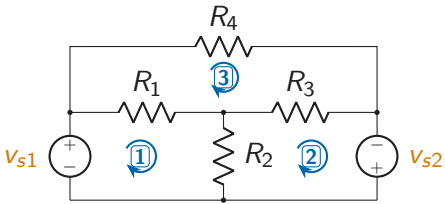



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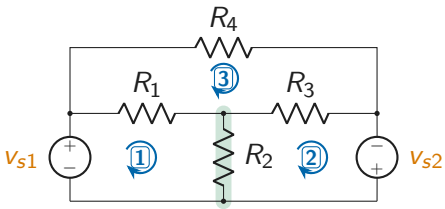

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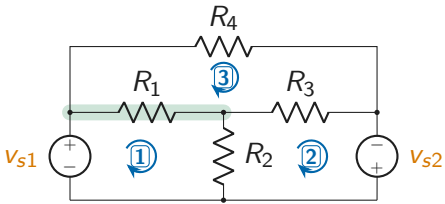

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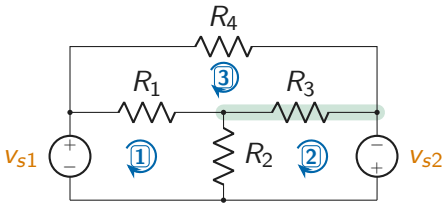

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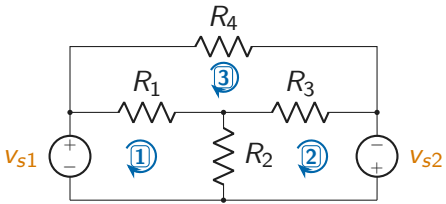

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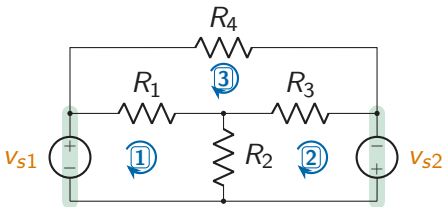
**Diagonal** $(k,k)$: $+\sum R$ around mesh $k$

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**Right side** $k$: $\sum v_s$ rising along mesh $k$

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

# Writing $Z_M$ directly from the circuit




Read $Z_M$ from the circuit?

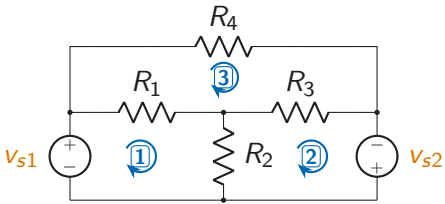
**Diagonal** $(k,k)$: $+\sum R$ around mesh $k$

**Off-diagonal** $(k,j)$: $-\sum R$ shared by $k$, $j$

**Right side** $k$: $\sum v_s$ rising along mesh $k$

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}$$

# Writing $Z_M$ directly from the circuit




Read $Z_M$ from the circuit?

**Diagonal** $(k,k)$: $+\sum R$ around mesh $k$

**Off-diagonal** $(k,j)$: $-\sum R$ shared by $k$, $j$

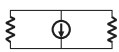
**Right side** $k$: $\sum v_s$ rising along mesh $k$

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}$$

Same as KVL ✓ $Z_M$ is symmetric

# Roadmap









**Voltage sources** ✓

the procedure, on an example ✓

matrix form ✓

**Current sources** 2

**on one mesh only**

between two meshes

**Dependent sources** 3

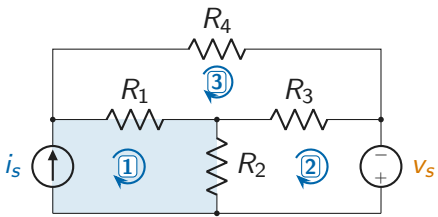
a dependent current source

**Node or mesh?** 4

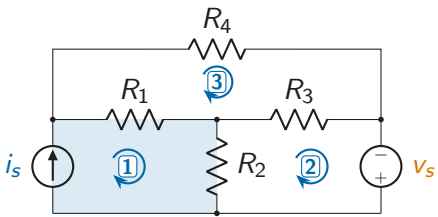
the same circuit both ways

# Example 12.2: a current source on one mesh only

$i_{M1} = i_s$: known



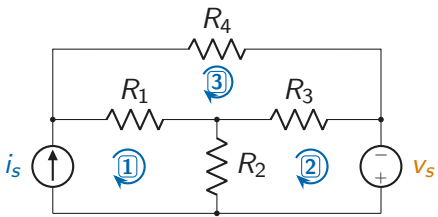
# Example 12.2: a current source on one mesh only




$i_{M1} = i_s$: known

No KVL around mesh 1. Unknowns: $i_{M2}$, $i_{M3}$

# Example 12.2: a current source on one mesh only

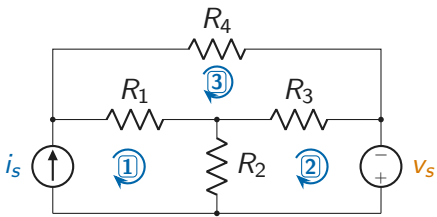



$i_{M1} = i_s$: known

No KVL around mesh 1. Unknowns: $i_{M2}$, $i_{M3}$

**Mesh 2:** $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

# Example 12.2: a current source on one mesh only




$i_{M1} = i_s$: known

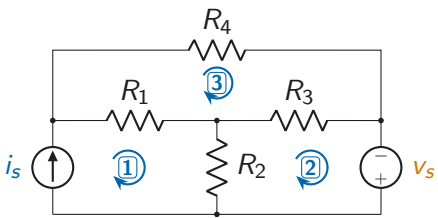
No KVL around mesh 1. Unknowns: $i_{M2}$, $i_{M3}$

**Mesh 2:** $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

**Mesh 3:**

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

# Example 12.2: a current source on one mesh only




$i_{M1} = i_s$: known

No KVL around mesh 1. Unknowns: $i_{M2}$, $i_{M3}$

**Mesh 2:** $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

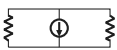
**Mesh 3:**

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}$$

# Roadmap










**Voltage sources** ✓

the procedure, on an example ✓

matrix form ✓

**Current sources** 2

on one mesh only ✓

**between two meshes**

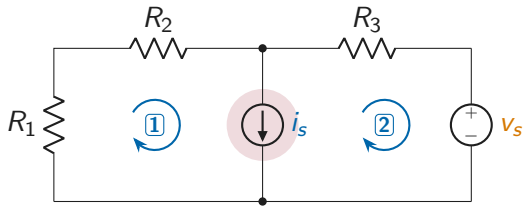
**Dependent sources** 3

a dependent current source

**Node or mesh?** 4

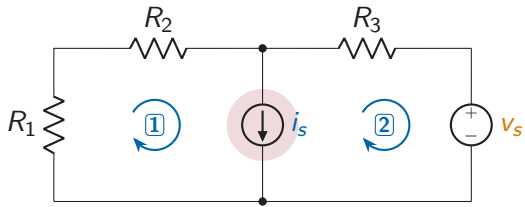
the same circuit both ways

# Example 12.3: a current source between two meshes



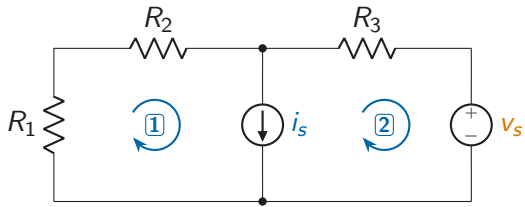

- KVL around mesh 1: voltage across $i_s$?

# Example 12.3: a current source between two meshes



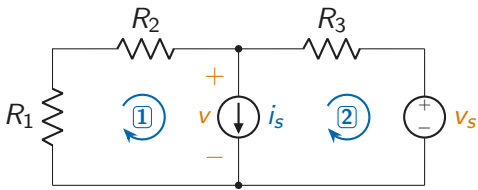

- KVL around mesh 1: voltage across $i_s$?
- Not given by $i_{M1}$, $i_{M2}$: a current source fixes $i$, not $v$

# Example 12.3: a current source between two meshes



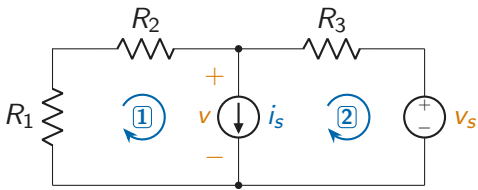

- KVL around mesh 1: voltage across $i_s$?
- Not given by $i_{M1}$, $i_{M2}$: a current source fixes $i$, not $v$
- Method I: its voltage as an unknown  Method II: supermesh

# Method I: the voltage as an unknown




New unknown: $v$

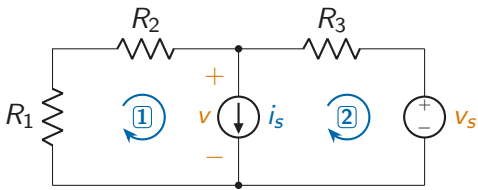
# Method I: the voltage as an unknown




**Mesh 1:** $(R_1 + R_2)\, i_{M1} + v = 0$

New unknown: $v$

# Method I: the voltage as an unknown

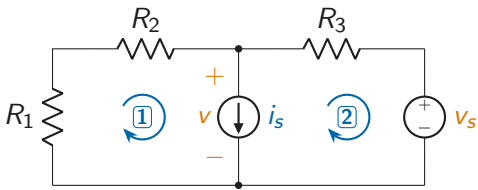



**Mesh 1:** $(R_1 + R_2)\, i_{M1} + v = 0$

New unknown: $v$

**Mesh 2:** $R_3\, i_{M2} + v_s - v = 0$

# Method I: the voltage as an unknown



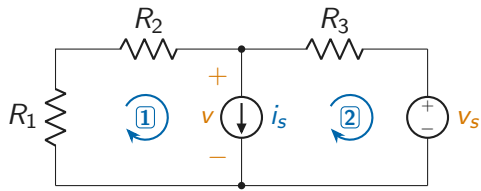

**Mesh 1:** $(R_1 + R_2)\, i_{M1} + v = 0$

New unknown: $v$

**Mesh 2:** $R_3\, i_{M2} + v_s - v = 0$

**Constraint:** $i_{M1} - i_{M2} = i_s$

# Method I: the voltage as an unknown

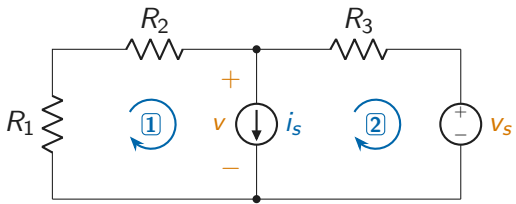



$$\begin{bmatrix} R_1+R_2 & 0 & 1 \\ 0 & R_3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ -v_s \\ i_s \end{bmatrix}$$

New unknown: $v$

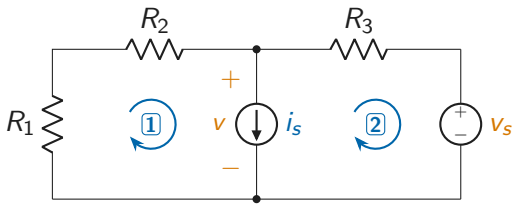
3 unknowns, 3 equations

# Method II: the supermesh




Can we get rid of $v$?

# Method II: the supermesh

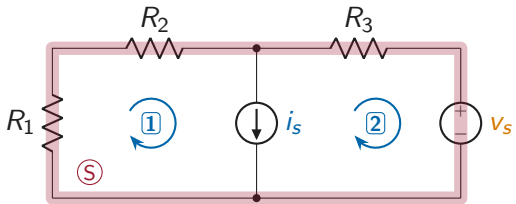



Can we get rid of $v$?

Add KVL of meshes 1 and 2:

$v$ cancels

# Method II: the supermesh

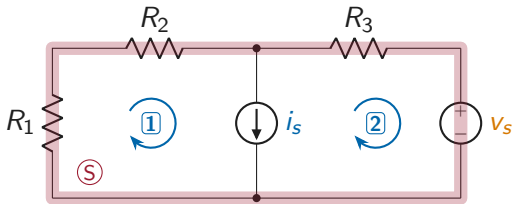



Can we get rid of $v$? = KVL around the outer loop (S)

Add KVL of meshes 1 and 2:

$v$ cancels

# Method II: the supermesh




Can we get rid of $v$?

Add KVL of meshes 1 and 2:

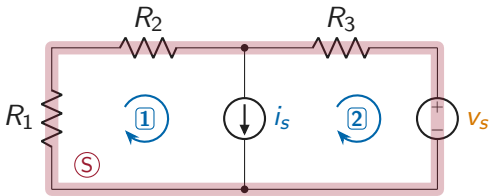
$v$ cancels

$=$ KVL around the outer loop (S)

**Supermesh**

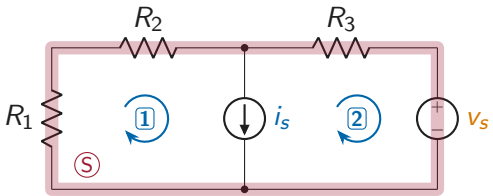
Two meshes as one, skipping the current source

# Method II: the equations



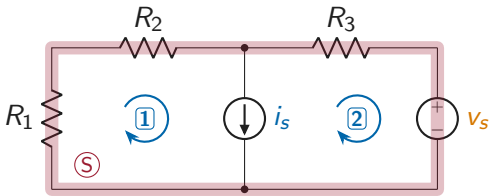

KVL around the supermesh?

# Method II: the equations




**Supermesh:** $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

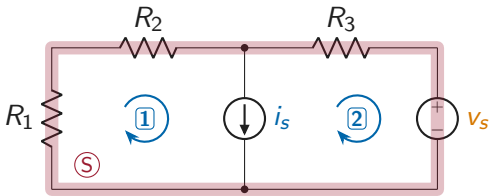
# Method II: the equations




**Supermesh:** $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

**Constraint:** $i_{M1} - i_{M2} = i_s$

# Method II: the equations

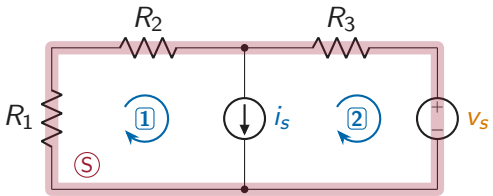



**Supermesh:** $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

**Constraint:** $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

# Method II: the equations



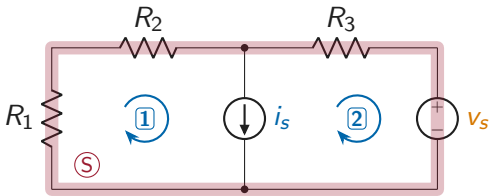

**Supermesh:** $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

**Constraint:** $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

2 unknowns (one fewer)

# Method II: the equations




**Supermesh:** $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

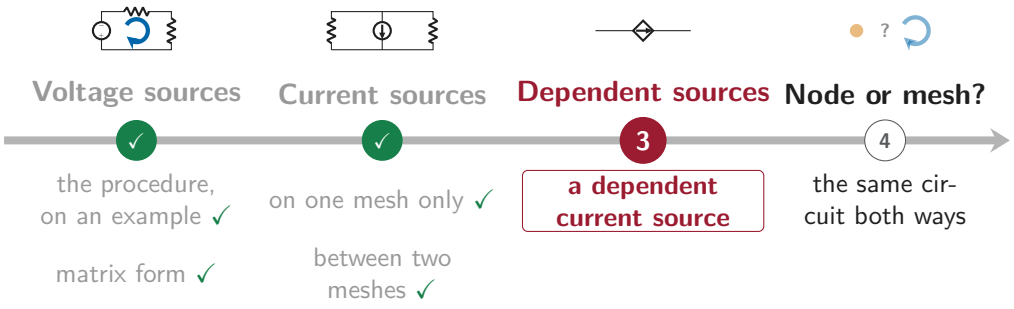
**Constraint:** $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

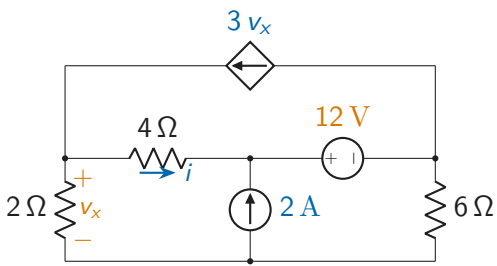
2 unknowns (one fewer)

Not symmetric

# Roadmap

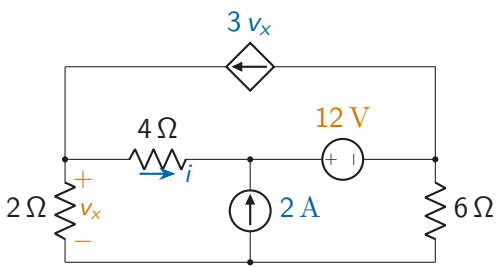

Voltage sources
Current sources
Dependent sources
Node or mesh?
✓
✓
3
4
the procedure, on an example ✓
matrix form ✓
on one mesh only ✓
between two meshes ✓
a dependent current source
the same circuit both ways

# Example 12.4: a dependent current source




Find $i$ of the 4 Ω.

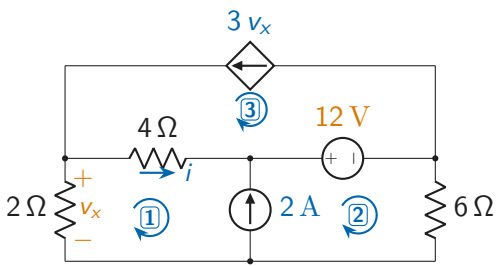
# Example 12.4: a dependent current source




Find $i$ of the 4 Ω.

Treat $3\,v_x$ as a current source

# Example 12.4: a dependent current source

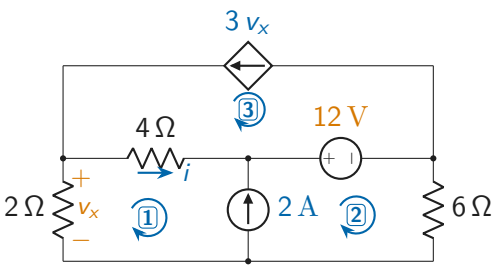



Find $i$ of the 4 Ω.

Treat $3\,v_x$ as a current source

Unknowns: $i_{M1}$, $i_{M2}$, $i_{M3}$

# Example 12.4: a dependent current source



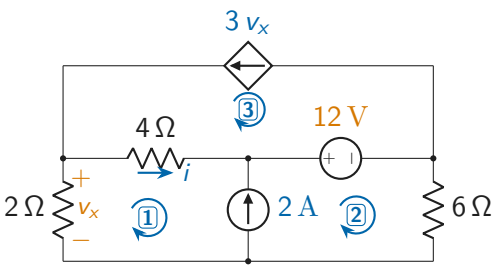

Find $i$ of the 4 Ω.

Treat $3\,v_x$ as a current source

Unknowns: $i_{M1}$, $i_{M2}$, $i_{M3}$

2 A between meshes 1, 2: supermesh

# Example 12.4: a dependent current source




Find $i$ of the 4 Ω.

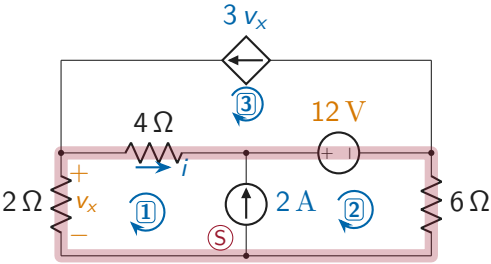
Treat 3 $v_x$ as a current source

Unknowns: $i_{M1}$, $i_{M2}$, $i_{M3}$

2 A between meshes 1, 2: supermesh

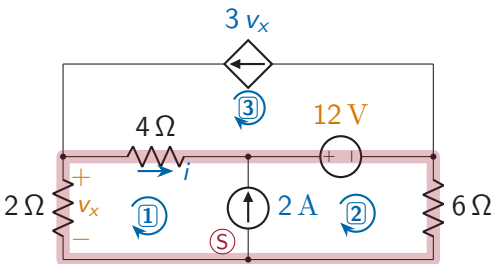
3 $v_x$ on mesh 3 only: fixes $i_{M3}$

# Example 12.4: the equations




KVL around the supermesh?

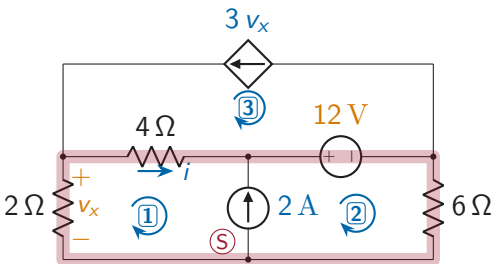
# Example 12.4: the equations




**Supermesh:**

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

# Example 12.4: the equations

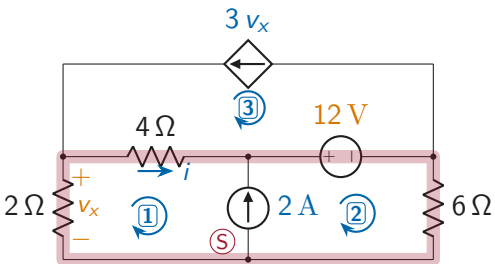



**Supermesh:**

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

2 A **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

# Example 12.4: the equations



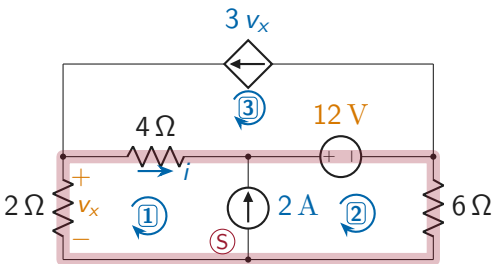

**Supermesh:**

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

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**Dependent source:** $i_{M3} = -3\,v_x$

# Example 12.4: the equations




**Supermesh:**

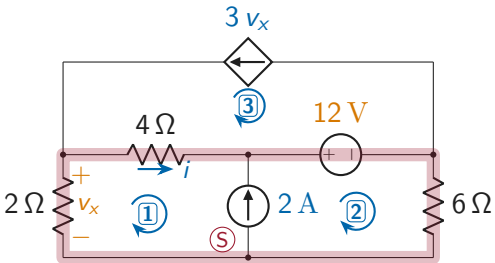
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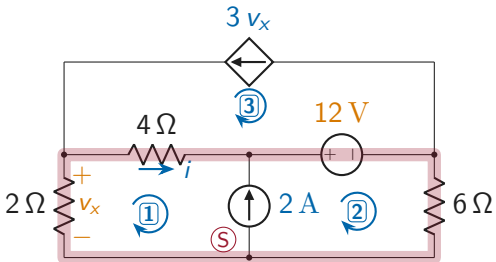
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$v_x$ in mesh currents?

$v_x = -2\,i_{M1}$

**Constraint:** $i_{M3} = 6\,i_{M1} \quad (3)$

# Example 12.4: matrix form

(1), (2), (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

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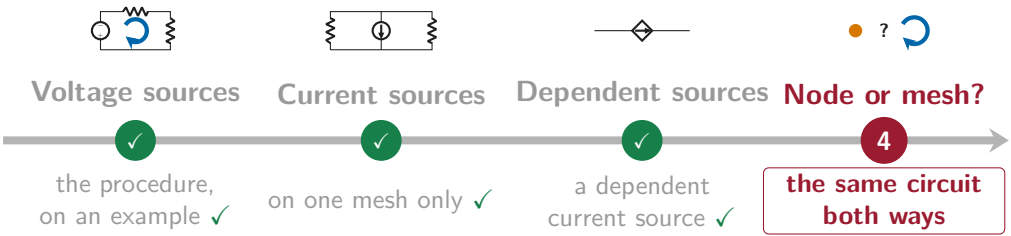
- Like an independent source of the same kind
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**Remark**

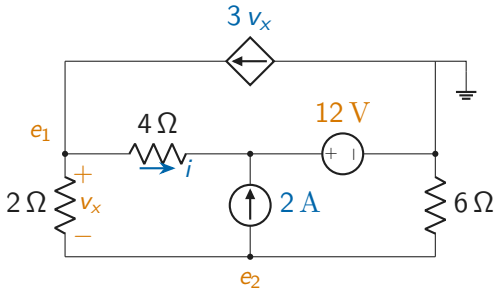
Dependent current source on mesh $k$ only:

no KVL around mesh $k$; write the constraint instead

# Roadmap

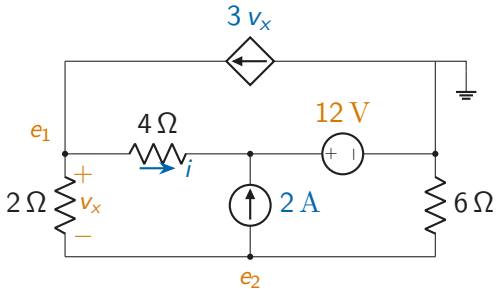

Voltage sources
Current sources
Dependent sources
Node or mesh?
?
4
the procedure, on an example ✓
matrix form ✓
on one mesh only ✓
between two meshes ✓
a dependent current source ✓
the same circuit both ways

# Example 12.4 with node voltages




Same circuit, node voltages?

# Example 12.4 with node voltages

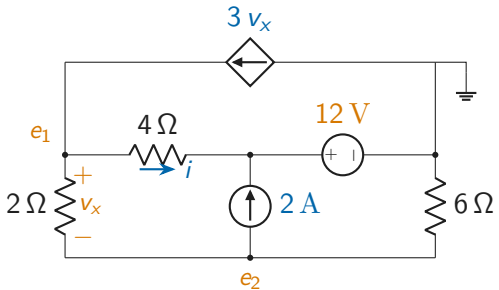



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Middle node: 12 V known

Unknowns: $e_1$, $e_2$

# Example 12.4 with node voltages



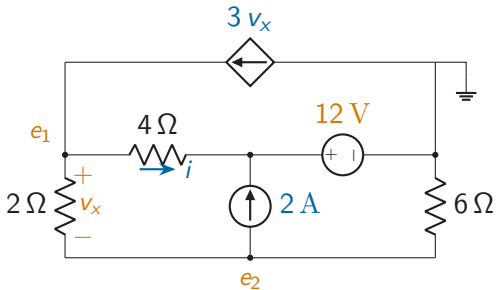

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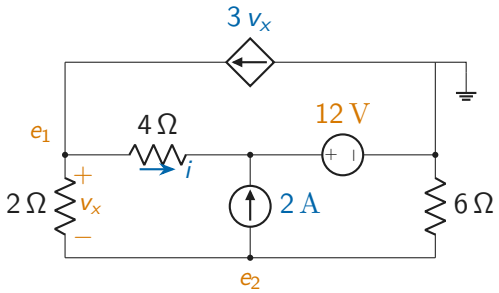
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# Example 12.4 with node voltages: matrix form

**First** plug $v_x = e_1 - e_2$ into node 1:

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$$i = \frac{e_1 - 12}{4} = \frac{-28 - 12}{4} = -10\,\mathrm{A} \quad \checkmark$$

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Example 12.4: which needs fewer equations?

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$\ell = 3$, 2 current sources

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## Node or mesh?

Example 12.4: which needs fewer equations?

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$3 - 1 = 2$ equations

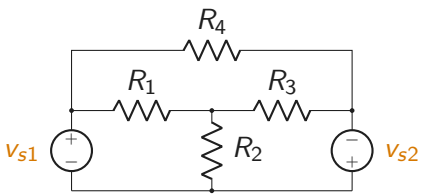
**Tip.** Count first. Mesh needs a planar circuit.

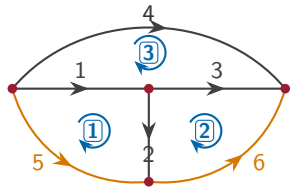
# Summary

| In the circuit | What to do |
|---|---|
| resistors, voltage sources | $Z_M \mathbf{i}_M = \mathbf{v}_M$, by the rules; symmetric |
| current source on one mesh | $i_{Mk}$ known; no KVL around mesh $k$ |
| current source between meshes | Method I or supermesh, plus constraint |
| dependent source | control in mesh currents: constraint |
| node or mesh? | count; mesh needs planar |

Then $\mathbf{i} = M^T \mathbf{i}_M$ and the element laws

# $Z_M$ from the graph: the mesh matrix

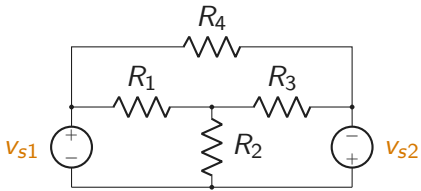


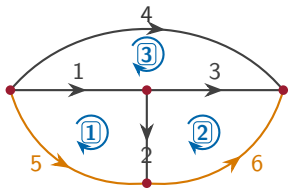




1–4: $R_1, \ldots, R_4$; 5, 6: $v_{s1}$, $v_{s2}$ ($+$ to $-$)

Mesh matrix $M$?

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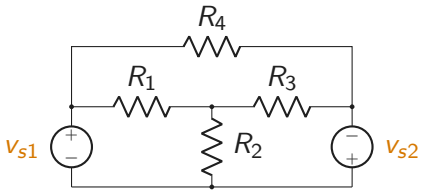
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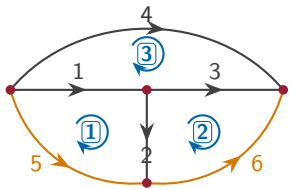
One row per inner mesh

$+1$ along, $-1$ against

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

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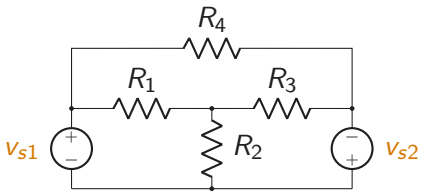
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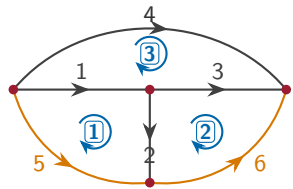
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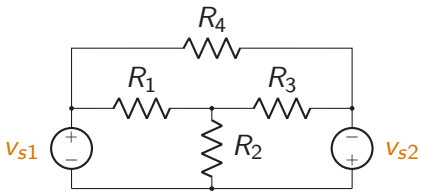
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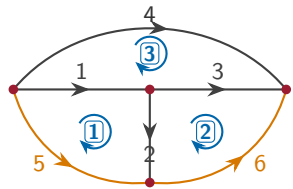
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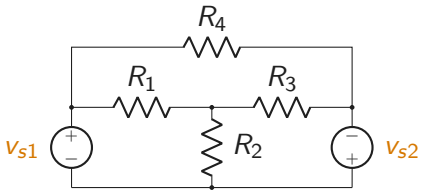
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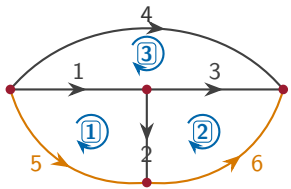
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$M_R$: resistors $M_S$: sources

# $Z_M$ from the graph: KVL, KCL and Ohm's law in blocks

Resistors first, then voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}$$

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**KCL:**

$$\begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$$

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**Ohm:** $\mathbf{v}_R = R\,\mathbf{i}_R, \quad R = \mathrm{diag}(R_1, \ldots, R_{b_R})$

# $Z_M$ from the graph: the general result

Ohm and KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

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$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

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Ohm and KCL into KVL:

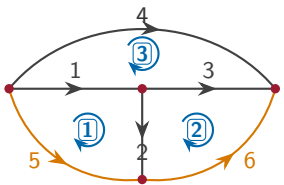
$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

$Z_M^T = M_R\, R^T M_R^T = Z_M$: always symmetric

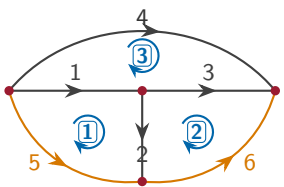
# $Z_M$ from the graph: Example 12.1, the resistor currents

$M_R^T$: columns 1–4 of $M$, as rows



# $Z_M$ from the graph: Example 12.1, the resistor currents

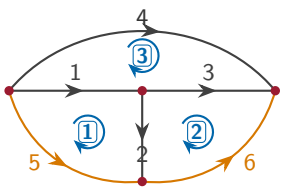
$M_R^T$: columns 1–4 of $M$, as rows




$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}$$

# $Z_M$ from the graph: Example 12.1, the resistor currents

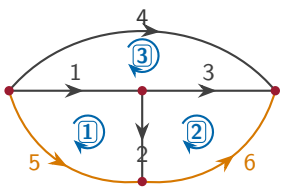
$M_R^T$: columns 1–4 of $M$, as rows




$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

# $Z_M$ from the graph: Example 12.1, the resistor currents

$M_R^T$: columns 1–4 of $M$, as rows

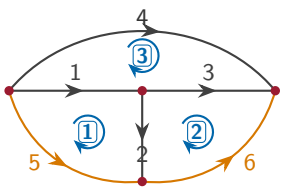



$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Along minus against

# $Z_M$ from the graph: Example 12.1, the resistor currents

$M_R^T$: columns 1–4 of $M$, as rows




$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Along minus against

Ohm: $v_{R_k} = R_k i_{R_k}$

# $Z_M$ from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

# $Z_M$ from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}$$

# $Z_M$ from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

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# $Z_M$ from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

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$$-M_S \mathbf{v}_S = - \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}$$

# $Z_M$ from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = -\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix} = \mathbf{v}_M \quad \checkmark$$

# $Z_M$ from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = -\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix} = \mathbf{v}_M \quad \checkmark$$

The rules = this product